

[10]
✓ 0s

▶ #How do we remove rows that failed the attention check?
#Dishant Sharma 12530084
import pandas as pd, numpy as np
df = pd.read_csv(r"/MRF_balance.csv")
df = df[df['tad'] == 1].copy()
df.reset_index(drop=True, inplace=True)
print(df.shape)

▼ ... (3, 12)



1] Os

```
#How do we select only the variables we need?
#Dishant Sharma 12530084
import pandas as pd, numpy as np
df = pd.read_csv(r"/MRF_balance.csv")
VARS = ['equitycapital', 'totalliabilities', 'totalassets']
df = df[VARS].copy()
print(df.head())
```

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...	equitycapital	totalliabilities	totalassets
0	839	8551	8551
1	839	8060	8060
2	839	8219	8219
3	1030	8865	8865
4	1030	8945	8945

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```
#How do we delete rows where more than 50% of values are missing?  
#Dishant Sharma 12530084  
import pandas as pd, numpy as np  
df = pd.read_csv(r"/MRF_balance.csv")  
thresh = int(0.5 * df.shape[1])  
df = df.dropna(thresh=thresh).copy()  
df.reset_index(drop=True, inplace=True)  
print(df.shape)
```

... (9, 12)

▶ #How do we fill missing numeric values with the column mean?
#Dishant Sharma 12530084
import pandas as pd, numpy as np
df = pd.read_csv(r"/MRF_balance.csv")
num = df.select_dtypes(include=[np.number]).columns
for col in num:
 df[col].fillna(df[col].mean(), inplace=True)
print(df.isnull().sum())

... year 0
equitycapital 0
Reserves 0
deposits 0
otherliabilities 0
totalliabilities 0
fixedassets 0
cwip 0
investments 0
otherassets 0
totalassets 0
tad 0
dtype: int64

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#Dishant Sharma 12530084

#How do we ensure all numeric columns have the correct data type?

```
import pandas as pd, numpy as np
```

```
df = pd.read_csv(r"/MRF_balance.csv")
```

```
num_cols = ['equitycapital', 'totalliabilities', 'totalassets']
```

```
for col in num_cols:
```

```
    df[col] = pd.to_numeric(df[col], errors='coerce')
```

```
print(df.dtypes)
```



```
... year                object
equitycapital          int64
Reserves               int64
deposits               int64
otherliabilities       int64
totalliabilities       int64
fixedassets           int64
cwip                  int64
investments            int64
otherassets            int64
totalassets            int64
tad                   int64
dtype: object
```

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```
#How do we remove duplicates and standardise column names?
#Dishant Sharma 12530084
import pandas as pd, numpy as np
df = pd.read_csv(r"/MRF_balance.csv")
df.drop_duplicates(inplace=True)
df.columns = (df.columns.str.strip().str.lower().str.replace(r'^a-z0-9+', '_', regex=True))
df.reset_index(drop=True, inplace=True)
print(df.columns.tolist())
```



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```
... ['year', 'equitycapital', 'reserves', 'deposits', 'otherliabilities', 'totalliabilities', 'fixedass
```

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```
#How do we get a summary of the dataset's structure?
#Dishant Sharma 12530084
import pandas as pd, numpy as np
df = pd.read_csv(r"/MRF_balance.csv")
print(df.shape)
df.info()
print(df.describe())
```

```
... (9, 12)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   year                  9 non-null     object
1   equitycapital          9 non-null     int64
2   Reserves              9 non-null     int64
3   deposits              9 non-null     int64
4   otherliabilities      9 non-null     int64
5   totalliabilities      9 non-null     int64
6   fixedassets           9 non-null     int64
7   cwip                  9 non-null     int64
```

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#How do we compute mean, median, min, max and std deviation?

#Dishant Sharma 12530084

```
import pandas as pd, numpy as np
df = pd.read_csv(r"/MRF_balance.csv")
num = df.select_dtypes(include=[np.number]).columns
stats = pd.DataFrame({
    'Mean':df[num].mean(), 'Median':df[num].median(),
    'Min':df[num].min(), 'Max':df[num].max(),
    'Std':df[num].std()})
print(stats.round(2))
```

	Mean	Median	Min	Max	Std
equitycapital	966.78	1030.0	839	1033	95.84
Reserves	2334.33	2107.0	1012	4309	1211.88
deposits	4260.44	4196.0	3378	5335	602.86
otherliabilities	3370.11	2661.0	1779	6415	1584.81
totalliabilities	10932.00	8945.0	8060	17091	3328.41
fixedassets	2236.11	2152.0	1879	3059	435.37
cwip	23.11	20.0	2	53	17.32
investments	2.56	3.0	0	5	2.07
otherassets	8670.00	7021.0	5884	14014	2944.27
totalassets	10932.00	8945.0	8060	17091	3328.41
tad	0.33	0.0	0	1	0.50

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```
#How do we create new derived columns from existing ones?  
#Dishant Sharma 12530084  
import pandas as pd, numpy as np  
df = pd.read_csv(r"/MRF_balance.csv")  
df['debt_to_equity'] = df['deposits'] / df['equitycapital']  
df['asset_util'] = df['totalliabilities'] / df['totalassets']  
df['invest_ratio'] = df['investments'] / df['totalassets']  
print(df[['debt_to_equity', 'asset_util', 'invest_ratio']].round(3))
```

```
...  debt_to_equity  asset_util  invest_ratio  
0      5.001      1.0      0.0  
1      4.657      1.0      0.0  
2      4.480      1.0      0.0  
3      3.280      1.0      0.0  
4      3.912      1.0      0.0  
5      4.170      1.0      0.0  
6      4.355      1.0      0.0  
7      4.810      1.0      0.0  
8      5.165      1.0      0.0
```

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```
#How do we filter rows based on a condition?
#Dishant Sharma 12530084
import pandas as pd, numpy as np
df = pd.read_csv(r"/MRF_balance.csv")
high = df[df['totalassets'] > 200000]
print(high[['year', 'totalassets']])
both = df[(df['totalassets'] > 200000) & (df['tad'] == 1)]
print(both[['year', 'totalassets', 'tad']])
```

... Empty DataFrame
Columns: [year, totalassets]
Index: []
Empty DataFrame
Columns: [year, totalassets, tad]
Index: []

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```
#How do we compute group-wise statistics using groupby?  
#Dishant Sharma 12530084  
import pandas as pd, numpy as np  
df = pd.read_csv(r"/MRF_balance.csv")  
grp = df.groupby('tad')[['equitycapital', 'totalliabilities', 'totalassets']].mean().round(2)  
print(grp)
```

...	equitycapital	totalliabilities	totalassets
tad			
0	934.50	8892.17	8892.17
1	1031.33	15011.67	15011.67